

Rishabh Kumar

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EDUCATION

Vellore Institute of Technology
Bachelor of Technology in Computer Science and Engineering

Bhopal, India
Aug 2022 – May 2026

PROFESSIONAL EXPERIENCE

AI Research Engineer Intern
Planto.ai

Feb 2025 – Apr 2025
Remote

- Engineered the development of Python pipelines to scrape inference metrics, reducing manual reporting overhead by 20% and accelerating the team's evaluation cycle.
- Led the comparative benchmarking of 3 State-of-the-Art (SOTA) coding models (Copilot, StarCoder, GPT-4) across 4 enterprise codebases, measuring token latency and code acceptance.
- Informed architectural decisions by presenting technical benchmarks on model performance to the CEO by prepared 8+ technical strategy briefs, ensuring the selection of the optimal model for internal tooling.

PROJECTS

AutoForge : Self-Healing AI Project Generator
Python, LangChain, Gemini, Pydantic

Feb 2026– March 2026
([GitHub](#) | [Live](#))

- Built **AutoForge**, an autonomous AI software engineer that generates complete multi-file projects (frontend + backend) using **Pydantic-structured outputs, LangChain, and LLMs**.
- Designed a **self-healing execution pipeline** with Docker-based sandboxing that automatically runs, debugs, and fixes code via recursive error feedback loops.
- Implemented **context-aware incremental updates & evaluation engine**, enabling scalable project modifications with relevance analysis and automated quality scoring.

Mental Health RAG System
FastAPI, Qdrant, Gemini, Hugging Face, MongoDB

Nov 2025 – Dec 2025
([GitHub](#))

- Orchestrated the end-to-end architecture of a hybrid RAG retrieval system, combining **Hugging Face embeddings** for local search with Gemini LLM for generation.
- Enforced strict safety guardrails by devising a 15-word sliding window check with vector similarity thresholds ($gt=0.85$), drastically reducing false negative rates.
- Achieved **sub-second latency** by implementing non-blocking **asynchronous workflows**, enabling the system to handle 50+ concurrent crisis alerts.

RepoChat
FastAPI, LangChain, ChromaDB, Gemini

Oct 2025 – Nov 2025
([GitHub](#))

- Built a LangChain-based RAG system** for conversational Q&A over GitHub repositories using ChromaDB and Gemini LLM, enabling semantic retrieval across large codebases.
- Implemented recursive document chunking with **800-token chunks and 100-token overlap**, improving retrieval accuracy for long source files.
- Developed FastAPI backend services for repository ingestion, vector indexing, and multi-turn chat interactions with conversation history management.
- Enabled semantic retrieval using ChromaDB with **top-k (k=4) vector search** for code-aware responses.

TECHNICAL SKILLS

Languages: Python, Java, SQL

Generative AI: RAG Architectures, LangChain, Hugging Face Transformers, PEFT (LoRA), Prompt Engineering, PyTorch

Vector & Data Infra: Qdrant, ChromaDB, MongoDB, Vector Similarity Search

Backend , Frontend & DevOps: FastAPI, React.js, Docker

ACHIEVEMENTS & OPEN SOURCE

- Open Source Contribution:** Merged stability patches to `terminal-ai` (Python), implementing **request timeout handling** to prevent CLI hangs during high-latency API calls.
- Certification:** Oracle Cloud Infrastructure (OCI) Certified **Generative AI Professional** (2025).
- Hackathon:** Secured Top 500 rank among 3000+ teams in HackOnBlocks.